

Programming languages

Why do we have many programming languages ?

لانه كل لغة برمجة الها ميزتها و لغات البرمجة متنوعة و مختلفة و كل لغة الها خصائصها و عيوبها

. ما في لغة برمجة عاطلة كل لغة الها وظيفتها و هدفها

Which language is better ?!

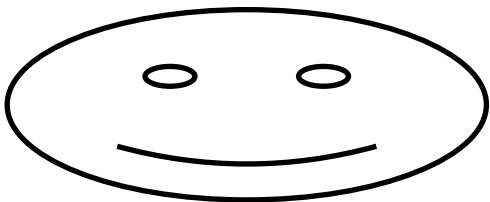
This is incomplete question !!

It depends on your need

و شو بدي اعمل ؟

You can use many programming languages at the same time.

ممکن ابرمج برنامج ب اكثر من لغة لانه كل لغة برمجة الها ميزاتها و كل لغة الها استخداما حسب الهدف يلي بدي استخدمها عشانه و انا كبرمج موضوع لغات البرمجة بالنسبة الي ما هي الا ادوات صغيرة بستخدمها حتى احقق هدف معين في برامجي



**Programming is important
not the language**

تعلم برمجة و ب امكانك استخدام اي لغة من لغات البرمجة لانه البرمجة
واحد بينما لغات البرمجة متعددة و ما هي الا ادوات

No / stop Racism (عنصرية) in programming languages

القاعدة العامة —<

كل لغات البرمجة رائعة و كلها ممكن استفيد منها و بستخدم يلي بتفيدني
حتى احقق الهدف المطلوب

How long it takes to learn a programming language ?!

Less than a month's time

لكن البرمجة نفسها على الاقل بدها سنتين

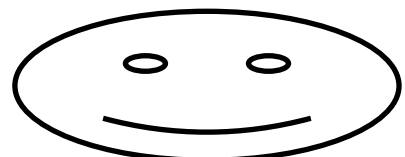
Learning a programming language its 5% of learn
programming

Understand the difference between programming &
programming language

Programming NOT EQUALS programming language

Programming language are only tools

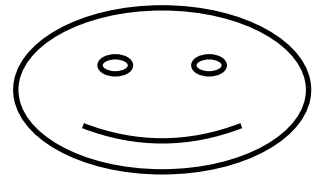
—> mastering them alone will not make you a good
developer



- Mastering programming language is only 5% of learning programming >-<
- Don't waste your time to learning more than one programming language >-<
- if you master 1 programming language you can learn any other language in less than month's time
- all programming languages share the same concepts , the only difference is syntax / tools

· *Focus on learning “programming”*
NOT programming languages

- what is programming ?!



Programming is

A process of creating a set of instructions that tell a computer how to perform a task

It all depends on :-

Logic / analysis / thinking / ability to find solutions

Regardless of any programming language

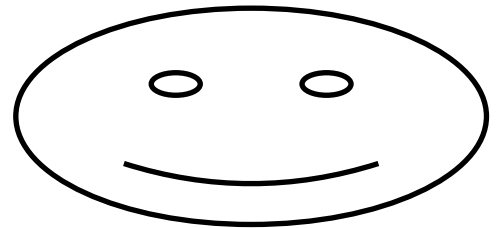
. البرمجة هي (logic) \ قدرتي على حل المشكلة و تحليل المشكلة و
تجزئتها الى مشاكل اصغر حتى اقدر القى الها حل و هي مجموعة من
الخطوات (process)

- Therefore mastering programming languages alone will not make you a good developer

You should Focus to learn “programming” rather than only learning “programming languages”

We agreed to focus on improving your —>

- 1) logic
- 2) analysis
- 3) thinking
- 4) ability to find solutions



Only using one programming language

But how & which language ??

Never ever start with a modern programming languages!!

Start with —> C++ or JAVA **why??**

Because —> it's a object oriented programming

The better start with C++ —> because it support

- 1) procedural programming
- 2) object oriented programming

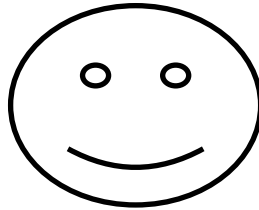
While java —>

Just support object oriented programming

لماذا يجب ان تبدأ بتعلم ++C

Why to start with C++?

C ++ = salary ++



للتذكير <—

لغات البرمجة هي مجرد ادوات يستخدم يلي بدي اياه منها ب
المكان يلي بدي اياه حتى احقق الهدف المطلوب

What is C ++ ?

C++ is a

- —> cross platform <—
- —> Portable language <—
- that can be used to create

cross platform &
Portable language
mean -->

بمعنى البرنامج يلي بكتبه
بهاي اللغة بقدر انزله على
اي (OS)

high performance applications

- That can be **adapted** to (((**multiple platforms**)))
- Is fun & easy to learn
- —> is the mother of all languages <—

لانه اللغات الثانية بشكل مباشر او غير مباشر مبنية على C++

- It will force you learn basic programming concepts

فهم المفاهيم البرمجة ب الاجبار و بشكل كامل

- it **supports procedural & OOP programming**
- If you learn (C++) very well , you can learn any other programming language easily in less than a month's time / in almost not time

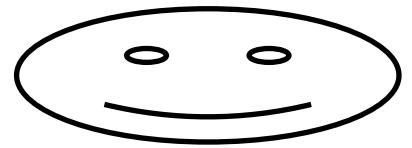
انواع لغات البرمجة :-

1) procedural programming (البرمجة الهيكلية)

2) object oriented programming (البرمجة الشيئية)

لغة (C++) بتدعم النوعين و بشكل كامل و هاي من اهم مميزاتاها و هي لغة تأسيسية 100% و بتبنيلي اساس قوي

What / why C++ !?



Both C & C++ are considered Mid - Level languages that are close to hardware & are able to program & manipulate the underlying hardware resources

(C++) بتمتاز ب قدرتها على التحكم ب (memory)

Both C & C++ have full control on computer resources (hardware) such as memory ,

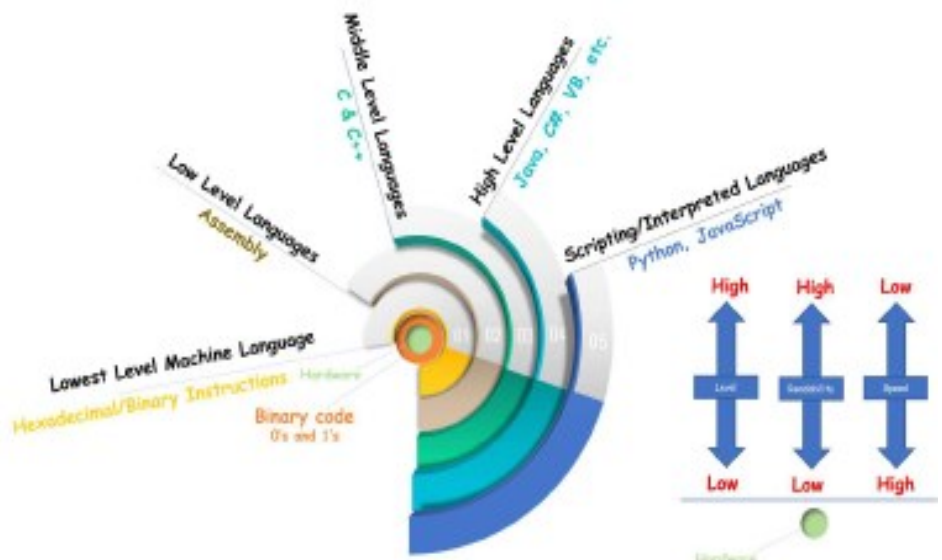
- no atomic memory management

ما فيهم ادارة للذاكرة بشكل (atomic) وتأثر على performance

Will I work C++?

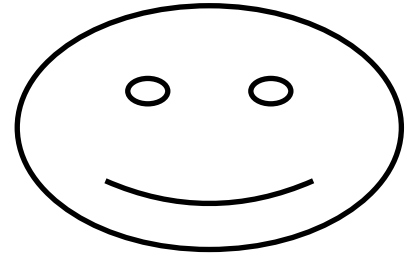
Yes and no

Because it still widely used for high performance systems



What can you do with C++?

- 1.To design OS (عمل OS كامل)
- 2.Desktop applications
- 3.Design compilers / programming languages
- 4.Design database
- 5.Engineering application
- 6.Game development
- 7.Medical applications
- 8.Desing web browsers
- 9.Embedded system



(الانظمة يلي ما الها شاشات مثل السيارات الحديثة و
(الطابعات وغيرها

- 10.Computation platforms
(الامور يلي بدها حسابات عالية)
- 11.Built enterprise software
- 12.IDEs (integrated development environment)

Real things done using C++ :-

1)operating systems

(windows) كلها معمولة على (C and C++)

Apple OS —> some of its parts written in C++

2) applications

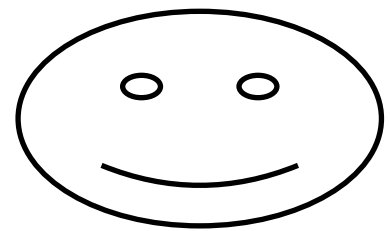
Amazon / Microsoft products / adobe photoshop
YouTube (core functions write in C++)

3) Browsers

Fire fox , chrome ,

4) Databases

MySQL



5) Gaming engines

برمجة الالعاب

(Graphics) هو اكثر اشئ بده —>

Power + processing + memory management

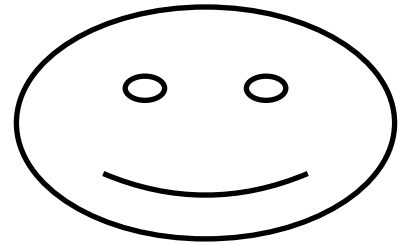
A game engine is a dedicated software or an architecture, or program that developers use for creating games

6) IDEs

Visual studio

7) enterprise software

Ex —> flight simulation
Radar processing



8) compilers & programming languages

(C#) مكتوبة على (C++)
(visual basic) مكتوبة على (C#)
(python) مكتوبة على (مكتباتها مكتوبة ب C++)

9) embedded system

مثل الانظمة يلي ب الطيارات و السيارات و الاجهزة الطبية

10) libraries

المكتبات عندي القدرة ابرمج مكتبة فيها (functions) اجزاء
ممکن استدعيها من اي لغة برمجة تانية و معظم المكتبات يلي بدها
(performance) عالي مكتوبة ب لغة (C and C++)

Most high-level machine language libraries use
C++ as backend

11) banking Applications

مثل (finacle) و غيرها

12) advanced computation and graphics

مثل maya , 3D graphics , animation , virtual reality

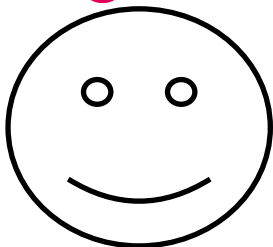
13) cloud / distributed systems

مثل (bloomberg)

C++ is **faster** than most of the other programming languages and also **supports multithreading with concurrency**

- **When we require very high-level mathematical computations, performance and speed** become important use C++ as their core programming language.

• Therefore , when you talk a bout -->
Performance , speed , cross platform , power ,
adaptable , multithreading with concurrency ,
high level mathematical computations ,
processing



It is C++

- why C++ not C ?!

Similarities -->

- Both the languages have a **similar Syntax/ Grammar/Code Structure**.
- The **compilation** of both the languages is **similar**
- **Both are mid level languages that are close to hardware** and are able to program and **manipulate** the underlying **hardware resources**

Differences -->

C -->

- developed 1969 to 1973
- Is a subset of C++
- It is procedural language

And remember !!!!!

IF you learn C++, you can learn any other programming language easily in almost not time.

C++ -->

- developed in 1979
- C++ superset of C
- It supports both language (procedural + OOP)

That's why you should learn C++ not C , it will take same efforts

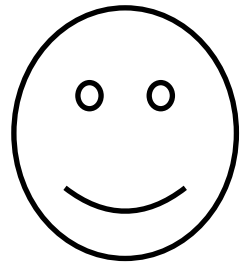
Why to start with C++ not java ??

Because -->

Java --> supports only OOP language

While

C++ --> supports both procedural and OOP



That's why you should start with C++ , you will learn all types of programming language